

# Operating System Architectures

## Fundamentals of Operating System Design

Prashant Jadav

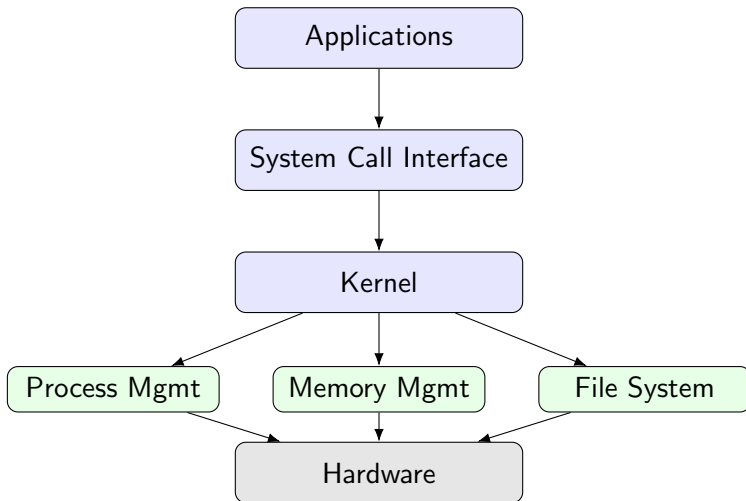
DEPSTAR, CHARUSAT

# Learning Outcomes

- Define OS architecture.
- Explain major kernel architectures.
- Compare their advantages and disadvantages.
- Identify architectures used by modern operating systems.

# What is an Operating System Architecture?

An operating system architecture defines how kernel components are organized and interact.

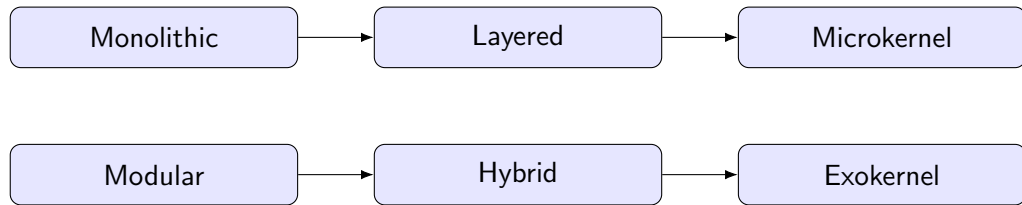


# Why Different Architectures?

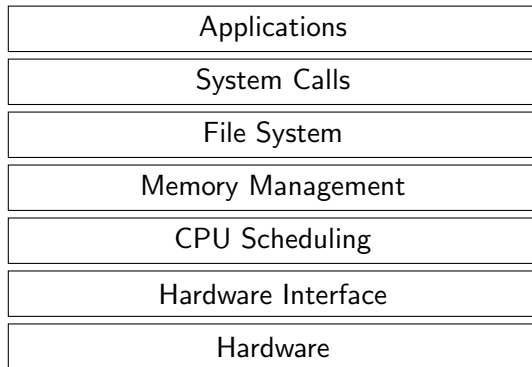
| Goal            | Motivation                         |
|-----------------|------------------------------------|
| Performance     | Faster execution                   |
| Reliability     | Fault isolation                    |
| Security        | Smaller trusted kernel             |
| Maintainability | Easier updates                     |
| Scalability     | Multi-core and distributed systems |

# Types of OS Architectures

- Monolithic Kernel
- Layered Architecture
- Microkernel
- Modular Kernel
- Hybrid Kernel
- Exokernel (advanced)



# Layered Architecture



Each layer uses services of the layer below.

# Microkernel

Kernel:

- IPC
- Scheduling
- Basic memory management

User Space:

- Drivers
- File Server
- Network Server

Examples: MINIX, QNX

Applications

Servers

Microkernel

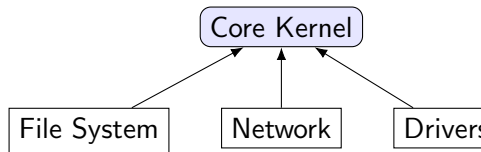
Hardware

# Modular Kernel

A modular kernel consists of a core kernel with dynamically loadable modules.

- Loadable Kernel Modules (LKMs)
- Runtime extensibility
- High performance
- Easy maintenance

Example: Linux





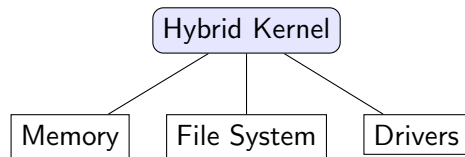
# Hybrid Kernel

Combines monolithic performance with microkernel modularity.

- Critical services in kernel mode
- Selected services modularized
- Balance of speed and reliability

Examples:

- Windows NT
- macOS (XNU)



- Minimal kernel
- Only resource protection and multiplexing
- OS abstractions implemented in library OS
- High flexibility and performance
- Primarily research-oriented

# Comparison of Architectures

| Architecture | Speed     | Reliability | Modularity | Example    |
|--------------|-----------|-------------|------------|------------|
| Monolithic   | High      | Medium      | Low        | Linux      |
| Layered      | Medium    | High        | High       | THE        |
| Microkernel  | Medium    | Very High   | Very High  | MINIX      |
| Modular      | High      | High        | High       | Linux      |
| Hybrid       | High      | High        | Medium     | Windows NT |
| Exokernel    | Very High | Medium      | High       | Xok        |

# Modern Operating Systems

| Operating System | Architecture           |
|------------------|------------------------|
| Linux            | Monolithic + Modules   |
| Windows NT       | Hybrid                 |
| macOS            | Hybrid (XNU)           |
| Android          | Linux-based Monolithic |
| MINIX 3          | Microkernel            |
| QNX              | Microkernel            |

- OS architecture defines kernel organization.
- Different architectures optimize different design goals.
- Modern operating systems mainly use modular or hybrid kernels.